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ABSTRACT

Purpose: Following sport-related concussions, early head impact exposure and premature return-to-sport are known to increase risk of repeat concussion in football athletes, yet athletes' true post-injury head impact exposure profiles (i.e., characteristics of recorded head impacts over a given time period) and biomechanical progression have not been explored. Accordingly, this study explored how head impact exposure in American college football athletes was altered during their return to sport from concussion, particularly within the same athletic season. **Methods:** This analysis compared daily volume of head impacts following concussion with pre-injury levels using head impact exposure profiles of fifty-two concussed collegiate football athletes from six NCAA Division I programs, and further compared these athletes to team- and position-matched controls to minimize season- or team-related factors. Additionally, this study provided an analysis of the possible association between duration of recovery and change in head impact exposure following concussion using continuous linear regression. **Results:** When comparing to pre-injury levels, 75% of concussed athletes reduced their head impact exposure in their immediate return-to-sport, while over 40% of concussed athletes did not reach their pre-injury level of head impact exposure at any point during the remainder of the concussion season segment. Furthermore, concussed athletes significantly decreased their head impact exposure over their immediate return-to-sport period when compared to team- and position-matched healthy, non-concussed athletes over the same time period. Finally, longer post-concussion recovery times were associated with larger decreases in head impact exposure after return-to-sport. **Conclusions:** This study provides evidence for a shift in head impact exposure after returning from concussion, seen most strongly in the immediate days after return-to-sport. These findings align with the recent shift toward more conservative post-concussion management seen across multiple sports and playing levels.

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Key Words: SPORT-RELATED CONCUSSION, SUBCONCUSSIVE HEAD IMPACTS, BIOMECHANICS, HEAD IMPACT SENSOR, CARE CONSORTIUM

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INTRODUCTION

Concussions have gained increased awareness in recent decades among clinicians, researchers, and the general public (1). While these injuries occur in any number of settings, including falls, motor vehicle collisions (2), and military blast trauma (3), sport-related concussions continue to receive the most attention. Estimates for sport-related concussion occurrence range from 1.6 to 3.8 million per year (4), yet its true occurrence is difficult to discern as sport-related concussions are frequently unreported, undiagnosed, or undocumented (5–7). As awareness of concussion has grown, increasing research in the field has made the pathophysiology and mechanisms of concussion clearer.

From a biomechanical standpoint, concussion has long been associated with high-magnitude head impacts producing significant head accelerations (8,9). However, the emergence of electronic sensors to measure head kinematics in the sports setting, such as the Head Impact Telemetry (HIT) system (10), has aided in better understanding the true biomechanics of concussion. Studies using these sensors revealed that the biomechanics associated with concussive impacts in the field were often modest compared to the experimental threshold of concussive linear and rotational accelerations derived from animal experimentation (11,12), suggesting that alternative mechanisms of concussion may also exist. Using data from the American football setting, Stemper and colleagues provided evidence that the cumulative effect of repetitive head impact exposure (HIE) is an additional possible mechanism of concussion (13). Repeated subconcussive head impacts may cause cumulative physiological and structural changes to the brain, leading to reduced tolerance and increased risk of further injury. Recent preclinical studies supported this assertion by demonstrating dose-dependent behavioral and pathological changes associated with repeated subconcussive head accelerations (14).

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Repetitive HIE may contribute to changes in structural, physiological, cognitive, and behavioral measures. Structural and physiological changes associated with repetitive HIE can be identified through imaging such as diffusion tensor imaging and functional MRI. Prior studies identified a correlation between elevated HIE and imaging changes (15–18). While not all imaging findings reflect irreversible changes, injurious impacts result in maladaptive structural changes that may produce cognitive and behavioral dysfunction. Unlike physical changes that have been extensively studied, prospective research surrounding cognitive and behavioral changes has been limited (19), with existing studies showing conflicting evidence for both the presence and absence of clinically-meaningful effects of HIE on neurological and cognitive measures (20–22). Nonetheless, studies have reported some evidence of impairments in reaction time, processing speed, memory and learning, balance, impulse control, and attention after one or two seasons of youth, high school, and college level athletics (20,23,24) and in different sports (25,26). However, despite the breadth of studies investigating these differences, strong evidence is lacking and confounding effects of age, sex, and other factors complicate the analysis. Ultimately, exposure to repetitive head impacts may disrupt the normal physiology and functioning of the brain, although continued investigation is necessary.

An important temporal component to the deleterious effects of repetitive HIE exists, particularly in the post-concussion period. Athletes who continued to accumulate HIE following a sport-related concussion were at risk for more severe acute concussion symptoms and longer recoveries than athletes who terminated participation in contact activities immediately following concussion (27,28). Participation in contact activity during the period after a concussion may expose the vulnerable brain to compounded neuropathophysiological processes. Thus, proper management of these athletes aimed at reducing HIE during this time period may be crucial in

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preventing intensified dysfunction in the brain. The cumulative result of this current research has shifted post-concussion management, as return-to-sport (RTS) time following concussion has more than doubled in recent decades (29). Athlete playing style and behavior after RTS following injury may be as important as medical management in safeguarding the vulnerable brain from HIE. Psychological factors and fear of reinjury may complicate athletes' RTS process as they re-initiate sporting activities (30). Whether an athlete's style of playing following RTS reflects the conservative shift seen in post-concussion management has not been explored.

A vast amount of research has been conducted on pre-injury biomechanics, HIE, and predisposing factors for concussion. However, knowledge of post-injury biomechanics is limited, particularly in short-term RTS within the same season. With a better understanding of athletes' recovery and RTS after concussion, insight can be gained on how clinical guidelines for concussion recovery promote or discourage the occurrence of HIE in football athletes. To that end, this is the first study to explore how American college football athletes alter playing style during their return from concussion, particularly within the same athletic season. Daily volume of head impacts following concussion was compared to pre-injury levels. Athletes were further compared to team- and position-matched uninjured controls, to determine if injured athletes had a greater change in HIE than athletes that were not concussed over the same time periods. Additionally, this study provided an analysis of the possible association between duration of recovery and change in HIE in RTS following concussion.

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METHODS

Population and Data Collection

This study used a subset of data from the NCAA-DoD Grand Alliance Concussion Assessment, Research, and Education (CARE) Consortium, including college football athletes from six NCAA Division I FBS programs: the United States Air Force Academy, the United States Military Academy at West Point, the University of California – Los Angeles, the University of North Carolina at Chapel Hill, the University of Wisconsin, and Virginia Tech. The study protocol was approved by the Institutional Review Board (IRB) at the Medical College of Wisconsin (MCW), with local sites utilizing a reliance agreement with the MCW IRB. The study also obtained approval from the US Army Human Research Protection Office (HRPO). Written informed consent was obtained from all participants. Each participating athlete was equipped with the helmet-based HIT System (10), which measures head accelerations using an array of linear accelerometers spring-mounted inside the football helmet and computes peak component and resultant linear accelerations of the head. Data acquisition triggered any time a single accelerometer exceeded a 9.6-g threshold. Only impacts with peak resultant linear acceleration greater than 10 g were included for analysis, as it was previously shown that head impacts below 10 g are commonly associated with non-impact activities such as running (31). Head impact data were recorded for all practice, scrimmage, and competition activities during the 2015-2019 football seasons, including preseason training camp, regular season practices and competitions, and spring practices.

Concussions were identified and diagnosed by team medical staff, according to a standardized protocol outlined by the CARE Consortium (32). Specifically, concussion was defined according to the consensus definition from the U.S. Department of Defense evidence-

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based guidelines initiative, which closely parallels the American Academy of Neurology definition: "a change in brain function following a force to the head, which may be accompanied by temporary loss of consciousness, but is identified in awake individuals with measures of neurologic and cognitive dysfunction (33)." NCAA return to sport guidelines as outlined in the Concussion Safety Protocol Checklist (34) did not have any major changes during the period of data collection for this study.

Analysis of HIE

Our analysis included athletes who sustained concussions in the 2015-2019 football seasons. For athletes with multiple concussions over this time period, only the first concussion was included. The analysis was further restricted to athletes with recorded contact days both before and after the concussion within the concussion season. This was done to establish both pre- and post-injury impact exposure profiles that consisted of the characteristics of recorded head impacts over a given time period. Given the differing nature of the preseason and regular season (e.g., competition-to-practice ratio, number of weekly contact days, etc.) which leads to differences in HIE (35,36), these season segments were assessed separately.

Data collected at several timepoints were used in this analysis: *Pre-Injury*, *First 7 Days Post-RTS*, and *>7 Days Post-RTS*. The *Pre-Injury* period served as an athlete's baseline for the concussion season segment (i.e., spring, preseason, or regular season). For preseason concussions, the *Pre-Injury* period included all practices and competitions from the initiation of activities for the concussion season up to but not including the date of concussion. For regular season concussions, the *Pre-Injury* period included all contact days beginning with the first regular season competition up to but not including the date of concussion. For spring concussions, the *Pre-Injury*

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period consisted of all contact days during the spring season prior to but not including the date of concussion. The day of concussion was not included in any period as practice/game participation and therefore HIE was likely altered by the occurrence of concussion.

Analysis of HIE during an athlete's return from concussion was divided into two distinct periods to capture differences in immediate and sustained responses – *First 7 Days Post-RTS* and *>7 Days Post-RTS*, respectively. Throughout an athlete's post-injury progress, the performance site's medical team managed the clinical care of the athlete based on current standards (37,38). Upon being diagnosed with concussion, the athlete was limited from both physical and cognitive activity until cleared to resume contact activity. Because of the phased return to sport protocol, this analysis used the Return to Sport date as the first date following concussion that head impacts were recorded using the HIT system. The *First 7 Days Post-RTS* period began on the first contact day (as determined by recorded head impacts in the HIT System) after the concussion date and included the subsequent six calendar days. The *>7 Days Post-RTS* period consisted of all remaining contact days after the *First 7 Days Post-RTS* period but still within the concussion season segment. The number of contact days in the >7 days post-RTS segment was not limited and included all days in that season segment as it was intended to capture the sustained behavior of the athlete after the initial return from concussion. It was necessary to restrict the *>7 Days Post-RTS* period to contact days within the same season segment, as HIE profiles differ between season segments (33,35). For example, if an athlete was injured in the preseason but did not return until the regular season, this athlete was not included in the analysis. For athletes who sustained a second concussion within the same season, time periods were adjusted only if the subsequent injury occurred within the same season segment as the first. For these athletes, the *First 7 Days Post-RTS*

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period was unchanged while the *>7 Days Post-RTS* period included all contact days after the *First 7 Days Post-RTS* period up to but not including the date of the second concussion.

The metric used to assess an athlete's post-injury response was "percent change in HIE." HIE was defined as the average number of daily head impacts, calculated by totaling all impacts within a given period (i.e., *Pre-Injury*, *First 7 Days Post-RTS*, *>7 Days Post-RTS*) and averaging across the number of contact days within that period. A percentage was then calculated to quantify the change in HIE during the RTS periods relative to *Pre-Injury* levels, according to Eq. 1. When considering a metric to be used, percent change in HIE better represented overall changes when comparing across all athletes, as absolute change in HIE was easily skewed by athletes with very low or very high levels of HIE.

$$\textbf{Eq. 1}: Percent\ Change = \frac{(HIE_{RTP} - HIE_{Pre-Injury})}{HIE_{Pre-Injury}} \times 100$$

Comparison to Team- and Position-Matched Controls

In addition to individual changes in HIE for the concussed athletes, percent change for each concussed player was compared to his position- and team-matched controls over the same post-injury periods. This was done to reduce the possibility that changes in HIE were associated with season- or team-related factors, as concussed athletes and their controls had very similar contact activity profiles. Each concussed player was matched with the study-enrolled players on his team that were in the same position group. Position groups included defensive linemen, linebackers, defensive backs, offensive linemen, tight ends, running backs, wide receivers, and quarterbacks. For example, percent change in HIE for a concussed offensive lineman at University X was compared to other non-concussed offensive linemen at the same University across the same pre-

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and post-injury periods, as seen in Figure 1. Practice days, times, and activities varied between programs and position groups; thus, this analysis eliminated much of that variability, resulting in observed changes that were more attributable to individual playing style.

A Rank Score was then calculated for each concussed athlete according to the method described in Stemper et. al, 2018 (13). This method takes into account the number of control athletes and generates a ranking for the concussed athlete, thus comparing his percent change to those of the matched control group. A Rank Score lower than the expected average of 0.5 indicates a concussed athlete had a larger percent change than his average matched control over the same time period. A Rank Score greater than 0.5 indicates a concussed athlete experienced a smaller percent change than his average matched control. Rank Scores for all concussed athletes were then grouped and statistically tested with a single sample *t*-test against the hypothetical Rank Score mean of 0.5, using a *p*-value of less than 0.05 as a statistically significant result. This analysis comparing concussed athletes' average Rank Score to the hypothetical Rank Score of 0.5 was performed separately for both the *First 7 Days Post-RTS* and *>7 Days Post-RTS* periods.

Length of Recovery versus Percent Change in HIE

An additional analysis was performed to explore the relationship between the amount of time an athlete required for recovery with his change in HIE once returning from injury. Recovery time was defined as the number of days between the concussion and the first day with recorded head impacts after the injury as determined by the HIT System. Percent change for both the *First 7 Days Post-RTS* and *>7 Days Post-RTS* periods was assessed. Using continuous linear regression and Pearson's correlation, statistically significant relationships were assessed using a *p*-value of less than 0.05.

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Analysis of Game Playing Opportunities

Changes in head impact exposure during the post-concussion period could be attributable to a variety of issues including changes in athlete playing style or altered playing opportunities. For example, athletes could be 'eased' back into game activities by participating in fewer snaps (i.e., playing opportunities) during immediate post-concussion period. Changes in playing opportunities were assessed by comparing the number of playing snaps per game that a player participated in both before concussion and during the post-concussion period. The number of offensive or defensive game snaps that each player participated in was obtained from a subscription-based website (Pro Football Focus, pff.com). Data obtained from pff.com were independently verified in a limited sample of games by visually tracking player participation using publicly available game videos. Analysis of playing opportunities was only conducted for concussed athletes in our sample that had logged offensive or defensive game snaps both before and after the date of their concussion.

The number of game snaps was logged for each game that concussed players participated in throughout the season in which they sustained their concussion. Offensive or defensive snaps were chosen based on a player's primary playing position. Special teams snaps were not included in the analysis. The total number of offensive or defensive snaps for the team was also logged for every game throughout that season. Athlete game participation was quantified as the percentage of offensive or defensive snaps that the player participated in for each game relative to the total number of offensive or defensive snaps for the team for that game. The average percentage of snaps that each concussed player participated in was calculated before and after their concussion. If the athlete sustained a concussion during a game, the percentage of snaps for that game was not included in the pre- or post-concussion average. Post-concussion participation was compared to

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pre-concussion participation to determine positive or negative changes in game participation for each concussed athlete following concussion.

RESULTS

Ninety-five (*n*=95) enrolled athletes across the six football programs included in this study sustained at least one concussion during the 2015-2019 seasons while wearing the HIT System. Of these athletes, 65 logged head impacts both prior to and after the concussion within the same concussion season. Fifty-two athletes remained after further selecting only athletes with HIE in both post-injury time periods (Figure 2). Among these athletes, concussions occurred across all segments of the college football season, including preseason (*n* = 17, 32.7%), regular season (*n* = 34, 65.4%), and spring season (*n* = 1, 1.9%). Given the format of the spring schedule more closely resembling that of the preseason (i.e., no competitions) (39), the single athlete with spring concussion was grouped with the preseason concussions when reporting results by season segment.

Pre-Injury HIE

Athletes sustained a wide range of daily head impacts during the *Pre-Injury* period. Average daily HIE for concussed athletes over the *Pre-Injury* period ranged from 1.5 to 28.6 impacts per contact day (median, 12.5 impacts; 95% CI, 10.9-14.5), as seen in Figure 3. Athletes who suffered preseason concussions experienced lower daily HIE (median, 9.9 impacts; 95% CI, 8.3-13.6) compared to athletes with regular season concussions (median, 13.2 impacts; 95% CI, 11.3-16.0) over fewer contact days prior to concussion (preseason: median, 4.0 preseason contact days; 95% CI, 3.0-5.3 vs. regular season: median, 14 regular season contact days; 95% CI, 13.5-23.1).

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Percent Change in HIE following Injury

Two post-injury time periods were chosen to capture both an athlete's initial response in HIE following RTS (i.e., *First 7 Days Post-RTS*), as well as changes in HIE after a period of stabilization (i.e., *>7 Days Post-RTS*). Athletes logged on average 3.6 ± 1.5 contact days in the *First 7 Days Post-RTS* period. Thirty-nine of 52 athletes (75.0%) exhibited a decrease in daily HIE from their *Pre-Injury* level during the *First 7 Days Post-RTS* period (median, 38.1% decrease; 95% CI, 5.0 to 42.0% decrease). Of the remaining 13 athletes, 1 athlete (1.9%) remained unchanged (i.e., percent change of zero) while 12 athletes (23.1%) increased their HIE from *Pre-Injury* levels over this period. Similar results were seen when assessing percent change over the remaining days within the season segment.

Athletes logged an average of 11.3 ± 10.1 contact days (preseason: 4.9 ± 3.0; regular season 14.8 ± 11.0) during the >7 Days Post RTS period. Thirty of 52 athletes (57.7%) exhibited a decrease in daily HIE from their *Pre-Injury* level during the *>7 Days Post-RTS* period (median, 7.8% decrease; 95% CI, 1.4 to 21.4% decrease). The remaining 22 athletes (42.3%) were unchanged or increased their HIE from *Pre-Injury* levels over this period. Absolute HIE for all athletes during both the *First 7 Days Post-RTS* and *>7 Days Post-RTS* is presented in Figure 3.

Of the 52 athletes in our study, 22 athletes (42.3%) decreased their HIE over the *First 7 Days Post-RTS* period and sustained a decrease over the *>7 Days Post-RTS* period. This group represents the lower left quadrant of Figure 4, indicating those concussed athletes who did not reach their *Pre-Injury* level of HIE at any point during the remainder of the concussion season segment.

To discern whether post-injury changes varied by the nature of the contact activity (i.e., practice vs. competition), contact days during the post-injury time periods were categorized as

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either practice or competition, and percent change in HIE was then calculated for each activity. Overall, percent change in HIE for both practices and competitions largely reflected the same trend seen when grouped together; larger decreases were seen immediately after RTS while milder decreases were seen over the longer-term period. When considering practices alone, 71.4% of athletes decreased their HIE during the *First 7 Days Post-RTS* period (median, 46.7% decrease; 95% CI, 4.0% increase to 39.5% decrease), and 57.1% of athletes decreased their HIE during the *>7 Days Post-RTS* period (median, 7.7% decrease; 95% CI, 0.4% increase to 21.8% decrease). When analyzing competitions alone, 78.6% of athletes decreased their HIE during the *First 7 Days Post-RTS* period (median, 34.3% decrease; 95% CI, 4.4 to 46.3% decrease), and 42.9% of athletes decreased their HIE during the *>7 Days Post-RTS* period (median, 7.2% increase; 95% CI, 15.5% decrease to 36.3% increase).

Comparison to Team- and Position-Matched Controls: Daily HIE

Control data could not be obtained for 7 athletes, and thus these athletes were omitted from this analysis. However, 44 of the remaining 45 athletes matched to two or more controls, with a similar number of controls for each time period (*First 7 Days Post-RTS*: range, 1-10 control athletes; mean ± S.D., 4.8 ± 2.5 vs. *>7 Days Post-RTS*: range, 1-11 control athletes; mean ± S.D., 4.9 ± 2.5).

When comparing the change in number of recorded head impacts per day from the *Pre-Injury* period to the *First 7 Days Post-RTS* period, twenty concussed athletes (44.4%) exhibited the largest percent decrease within their matched control group. This indicates that these 20 concussed athletes had a larger decrease in the number of daily recorded head impacts than all other athletes that were in the same position group on the same team during that period. An

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additional 7 concussed athletes recorded the second-largest percent decrease in the number of recorded head impacts per day within their matched control group, indicating that only one matched control athlete had a larger decrease in daily HIE during that period. Overall, this indicates that 60% of concussed athletes had the largest or second-largest percent decrease in daily HIE during the *First 7 Days Post-RTS* period among their matched control group, which had an average size of 5.8 athletes (Figure 5).

Similar results were seen over the *>7 Days Post-RTS* period. Eleven concussed athletes (24.4%) held the largest percent decrease in recorded head impacts per day relative to the *Pre-Injury* period within their matched control group during that period. This indicates that these 11 concussed athletes had a larger decrease in the number of daily recorded head impacts than all other athletes that were in the same position group on the same team during that period. An additional 7 concussed athletes logged the second-largest decrease in recorded head impacts per day within their matched control group, indicating that only one matched control athlete had a larger decrease in daily HIE during that period. Accordingly, this indicates that 40% of concussed athletes had the largest or second-largest percent decrease in daily HIE during the *>7 Days Post-RTS* period among their matched control group, which had an average size of 5.9 athletes (Figure 6). Additional player data was calculated to help characterize each concussion (see Supplemental Table 1, Supplemental Digital Content, which provides additional player information).

Statistical Analysis of Matched Control Groups using Rank Analysis

Using the Rank Score method, a ranking was calculated for each concussed player among his control group. Each concussed athlete's Rank Score was then collectively tested against the hypothetical average score of 0.5.

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Average Rank Score for concussed athletes during the *First 7 Days Post-RTS* and *>7 Days Post-RTS* period were 0.361 ($p < 0.01$) and 0.487 ($p = 0.74$), respectively. Given average Rank Scores of less than 0.5, concussed athletes as a group exhibited a rank closer to 1 among their matched controls over both post-injury periods. Statistical testing revealed a significantly larger decrease in HIE than controls during a concussed athlete's initial RTS (Figure 5). However, the larger decrease in HIE compared to controls seen over the remainder of the season segment was not significant (Figure 6).

Length of Recovery versus Percent Change in HIE

Length of recovery following concussion, measured by number of days to first recorded post-injury contact, varied among concussed athletes (median, 7.0 days; 95% CI, 6.8-9.9). When analyzed separately, preseason concussions displayed slightly shorter recovery times compared to regular season concussions (preseason: median, 6.5 preseason days; 95% CI, 4.3-7.1 vs. regular season: median, 9.0 regular season days; 95% CI, 7.6-11.9).

Five of six HIE percent change versus recovery time associations (Figure 7) displayed inverse correlations (correlation coefficient, r: range, 0.04 - -0.45). Three of these associations demonstrated correlation coefficients magnitudes greater than -0.1, suggesting that longer recovery times were associated with larger decreases in daily HIE following concussion. This finding was most evident for changes over the *>7 Days Post-RTS* period, as evidenced by significant p-values among regular season concussions and when grouping all concussions together.

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Analysis of Game Playing Opportunities

A total of sixteen concussed athletes participated in games prior to and after their concussion (Table 1). This cohort of athletes sustained concussion between the second and tenth weeks of the regular season, and participated in an average of 3.75 games (range: 1 to 7 games) prior to and 4.94 games (range: 2 to 10 games) post-concussion. Across the entire sample (pre- and post-concussion), athletes participated in an average of 41% (range: 4% to 100%) of their team's offensive or defensive snaps. Repeated measures ANOVA revealed that there were no significant differences between pre-concussion and post-concussion participation (p=0.581). On an individual basis, athletes had an average of 2% higher participation post-concussion than prior to concussion. Nine of sixteen athletes had the same or greater participation following concussion, leaving seven athletes that had reduced participation after their concussion. Only one athlete had greater than 12% reduction in post-concussion participation (-36%). That reduced participation was associated with a 69.6% reduction in head impact exposure during the first seven days following return-to-sport, and a 6.7% reduction in head impact exposure for the remainder of the season. Across the entire sample, changes in post-concussion game participation were not correlated to changes in head impact exposure either during the first seven days following return-to-sport (p=0.564) or for the remainder of the season (p=0.718).

DISCUSSION

This analysis explored football athletes' HIE biomechanics following RTS from concussion. This was the first study to characterize HIE within the same athletic season following concussion and compare to pre-injury levels. Overall, 75% of athletes decreased their HIE from pre-injury levels following RTS, while 57.7% decreased their HIE from pre-injury levels over the

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remainder of the season segment. Over 40% (22/52) of concussed athletes did not reach their pre-injury level of HIE at any point during the remainder of the concussion season segment. Importantly, this reduced HIE following RTS occurred while athletes did not have a remarkable change in participation from pre-injury levels. Furthermore, when compared to team- and position-matched control athletes, concussed athletes showed significantly larger decreases in HIE in their immediate RTS after concussion than non-concussed athletes over the same time period. The largest changes in HIE were seen in the immediate days following an athlete's RTS, while HIE drew closer to pre-injury levels over the remainder of the season. This trend was also seen when distinguishing practices from competitions and analyzing them as separate groups. This is in line with current protocols and RTS guidelines, which recommend a physical-exertion progression that begins only after the concussed athlete demonstrates a normal clinical examination or the resolution of concussion-related symptoms (37). These findings of reduced HIE over similar game participation levels seems to imply that there is a change in playing style after athletes return from concussion.

Additionally, when correlating recovery time with changes in HIE after RTS, longer recovery times were associated with larger sustained decreases in HIE. The causative link behind these associations was not identified in this study. However, RTS decision-making is a complex process, as it holds implications not only for the health and safety of the athlete, but also for the sports organization. As Yung and colleagues identified, recovery time may be influenced by a number of factors, including severity of concussion, as well as who is responsible for an athlete's medical decision-making, as coaches, athletes, family members, and trainers may each contribute different social pressures (40). In addition, psychological factors may play a role, as was seen in previous research tracking anterior cruciate ligament (ACL) reconstruction outcomes, and more

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recently in athletes‘ recoveries from sport-related concussion (SRC). Despite high rates of successful outcomes in terms of knee function, a relatively low rate of return to competitive sport participation was seen in studies tracking post-surgical ACL reconstruction outcomes, in part due to psychological factors such as lack of motivation and low confidence (41) Similarly, athletes suffering from heightened levels of anxiety or depression have longer recovery times from sport related concussion (42). Fear of reinjury, or kinesiophobia, is also commonly cited as a hindrance to successful postoperative return to pre-injury participation following both ACL reconstruction (41,43) and sport-related concussion . Individuals with high fear of re-injury experience more concussive symptoms (44) and take longer to achieve clinical recovery (30). Previous research on RTS following Achilles tendon rupture has shown that psychological readiness improves and kinesiophobia decreases with greater time from the injury (45). Our current data supports this line of thought within the setting of sport related concussion, as concussed athletes' HIE was reduced to a greater extent in the *First 7 Days Post-RTS* period, while HIE drew closer to P*re-Injury* levels over the *>7 Days Post-RTS* period, despite no significant changes in an athlete's game participation level post-concussion.

Despite recurrent concussion rates decreasing across all high school sports in recent years (46), fear of reinjury, or repeat concussion, may be a realistic concern in competitive football, thus contributing to reduced HIE upon RTS. Among the 52 athletes included in this study, 6 athletes sustained a repeat concussion during the remainder of their collegiate football career. Two of these athletes sustained their next concussion within the same season as their first concussion. Despite these athletes having a recovery period from their first concussion on average with other athletes in this study (6 and 15 days until their first post-concussion contact day), as well as exhibiting decreases in their HIE compared with *Pre-Injury* levels (-65.4% and -3.8% change over the *First*

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7 Days Post-RTS period, -32.6% and -53.1% change over the *>7 Days Post-RTS* period, respectively), a repeat concussion was sustained. While there is concern that premature RTS may increase the risk for subsequent concussion, this is not the only factor that predisposes to subsequent concussion.

While the majority of athletes decreased their HIE after RTS following a concussion, a notable amount of athletes increased their HIE from *Pre-Injury* levels. When reviewing individual athlete data, two athletes in particular had percent change in HIE at the extreme over the *First 7 Days Post-RTS* period, specifically both increases of 233.3%. Within the limitation of just two athletes, no distinct associations were seen to account for these disproportionate increases. For example, the two athletes attended different programs, played different positions, and had post-concussion recovery times that were consistent with the average among all concussed athletes. In theory, season segment may play a role, since athletes may be battling among teammates for a starting position while in the presesason. Alternatively, athletes participating in high physical contact sports such as football are known to exhibit aggressive behaviors and actions (47), and a prolonged period of recovery away from football after a concussion may predispose players to show extra aggression. Finally, social pressures provided by teammates, coaches, and family may influence an individual's performance upon RTS after concussion. Determining what may predispose an athlete to increase their HIE after concussion remains difficult and outside the scope of this study.

During the progression from injury to resumption of full contact activity, an athlete needed to be cleared for full participation in sports activities by medical personnel. The date this clearance was granted is considered the official RTS date. This date was available in the Head Impact Measurement database for just 26 athletes (50.0%). The best estimate for the RTS date for the

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remaining athletes was the first contact day following the concussion, as determined by the HIT System. Therefore, first contact after the injury was used as the starting date of the *First 7 Days Post-RTS* period for all athletes. When reviewing the 26 athletes with available RTS date, the first post-injury contact day occurred 2.6 ± 4.0 days prior to the RTS date, indicating athletes logged HIE prior to receiving full medical clearance. Considering this discrepancy in dates, percent change values over the *First 7 Days Post-RTS* period for these 26 athletes were shifted 15.9 ± 34.9% toward larger decreases when calculated using the first post-injury contact day, rather than the official RTS date. Likewise, percent change values over the *>7 Days Post-RTS* period were shifted 1.5 ± 23.3% toward larger decreases when calculated using the first post-injury contact day.

The findings of this study were restricted by the following limitations. This analysis only included subjects who recorded head impacts both before and after the concussion, within the same season segment. A number of factors could account for a lack of HIE within the same season segment. Athletes may not have worn the HIT system after their RTS. Alternatively, the concussion may have occurred too late in the preseason that the athlete's first post-injury impacts occurred in the regular season. By excluding athletes without post-injury impacts, we may miss a group of athletes with concussions so severe that they did not return in the same football season. Future analysis would benefit from exploring whether post-injury changes extend into future seasons or conversely whether prior high school concussions impact a collegiate athlete's playing style. Another limitation was the inconsistent availability of athletes' official RTS dates. As a result, the first post-injury contact day was used when determining the beginning of an athlete's full RTS, which resulted in minor shifts in post-injury percent change from values calculated using the official RTS date. Finally, no consideration was made for an athlete's roster status either before

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or after the concussion. For example, a starting athlete who was downgraded to reserve status after his RTS from concussion may experience reduced HIE as a result. However, our analysis of percentage of a team's snap counts for concussed athletes provided evidence that game playing opportunities didn't remarkably change post-concussion.

Another limitation is that head impacts in this database were not verified using game and practice videos. This has been an acknowledged limitation of our prior studies using this Head Impact Measurement database due to the large number of recorded head impacts (>420,000) across the duration of the study (examples (13,48–51)). The Head Impact Telemetry System has inherent inaccuracies similar to all types of instrumentation used to measure biomechanical signals. For example, the HIT System was shown to have higher accuracy in identification of the occurrence of head impacts with reduced accuracy in recording the magnitude and direction of impacts (52–54). Our analyses have attempted to limit the effect of these inaccuracies on the study results by reporting impact counts and performing comparative analyses, as we have done in the current analysis, which focused primarily on comparison of post-concussion to pre-concussion HIE in individual athletes. As such, our assumption was that many sources of HIT System inaccuracies were consistent over the duration of the analysis. This is a particularly strong assumption given that a primary source of inaccuracy is associated with helmet fit which would not be expected to vary throughout the season for a given athlete. Since these analyses focused on within-athlete comparisons, helmet fit would be a factor that would not change prior to and after concussion. Nonetheless, video verification of all impacts was not part of the current study, although some level of video verification remains a focus of continued efforts within our group.

This was the first study to explore the style of play and mechanical HIE profile of athletes following concussion, relative to pre-injury HIE levels. Three-quarters (75%) of concussed

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athletes reduced their HIE in their immediate RTS after concussion while 57.7% reduced HIE over the remainder of the season segment, despite no real change in their game participation level (i.e., percentage of team game snaps). Over 40% of concussed athletes did not reach their *Pre-Injury* level of HIE at any point during the remainder of the concussion season segment. Furthermore, when compared to position- and team-matched controls to reveal changes due to individual playing style, concussed athletes displayed reductions in HIE that were greater than non-concussed controls over their immediate RTS, with 60% of concussed athletes displaying either the largest or second-largest decrease in HIE among their matched control group. Finally, longer recovery times after concussion were associated with larger decreases in HIE over the remaining contact days within the season segment. Findings from this study provide evidence for a shift in playing style after returning from concussion, most notably in the immediate days after RTS.

Declarations

Ethics approval and consent to participate: The study protocol was approved by the Institutional Review Board (IRB) at the Medical College of Wisconsin (MCW), with local sites utilizing a reliance agreement with the MCW IRB. The study also obtained approval from the US Army Human Research Protection Office (HRPO). Written informed consent was obtained from all participants. *Availability of data and materials:* The datasets used and/or analyzed during the current study are accessible through a Data Sharing Request to the CARE Consortium. *Conflict of Interest:* The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute endorsement by the American College of Sports Medicine. The authors have no significant conflicts of interest to disclose. *Funding:* Research reported in this publication was supported by the National Institute of Neurological Disorders and Stroke of the National Institutes of Health under award number 1R01NS126449. This publication was also made possible with support from the Grand Alliance Concussion Assessment, Research, and Education (CARE) Consortium, funded in part by the

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FIGURE LEGENDS

Figure 1: HIE for position- and team-matched controls was measured across the same dates as the concussed athlete's *Pre-Injury*, *First 7 Days Post-RTS*, and *>7 Days Post-RTS* time periods. Triangles represent HIE for the concussed athlete, while circles represent HIE for two matched-control examples.

Figure 2: Athletes were selected according to established inclusion criteria.

Figure 3: Absolute HIE was measured for all individuals across three time periods: *Pre-Injury*, *First 7 Days Post-RTS*, and *>7 Days Post-RTS*. Thick lines indicate group medians. Circles indicate Preseason concussions, squares indicate Regular Season concussions, while triangles represent the combination of Preseason and Regular Season concussions. Athletes who sustained preseason concussions consistently exhibited lower median HIE on contact days than athletes who sustained regular season concussions across all three time periods.

Figure 4: Percent change from *Pre-Injury* average daily HIE was calculated over both the *First 7 Days Post-RTS* and *>7 Days Post-RTS* periods. Values for each individual athlete were plotted. The largest group of concussed athletes demonstrated decreases in daily HIE over both the *First 7 Days Post-RTS* and *>7 Days Post-RTS* periods.

Figure 5: Percent change (top) for each concussed athlete (triangles) was compared against team- and position-matched controls (circles) over the *First 7 Days Post-RTS* period. Rank Scores

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(bottom) were then calculated to further characterize how concussed individuals compared to uninjured controls, with a Rank Score closer to 0 indicating a larger magnitude decrease in percent change than the controls. Some control athletes with percent change greater than 400% are not shown on this figure but were included in the analysis.

Figure 6: Percent change (top) for each concussed athlete (triangles) was compared against team- and position-matched controls (circles) over the *>7 Days Post-RTS* period. Rank Scores (bottom) were then calculated to further characterize how concussed individuals compared to uninjured controls, with a Rank Score closer to 0 indicating a larger magnitude decrease in percent change than the controls. Some control athletes with percent change greater than 300% are not shown on this figure but were included in the analysis.

Figure 7: Percent change for both the *First 7 Days Post-RTS* and *>7 Days Post-RTS* periods was plotted against Time to Post-Injury Contact (measured in days), and linear regression lines were fit to the data. Longer recovery times were associated with larger decreases in daily HIE over the *>7 Days Post-RTS* period among regular season concussions and when grouping all concussions together.

Supplemental Digital Content

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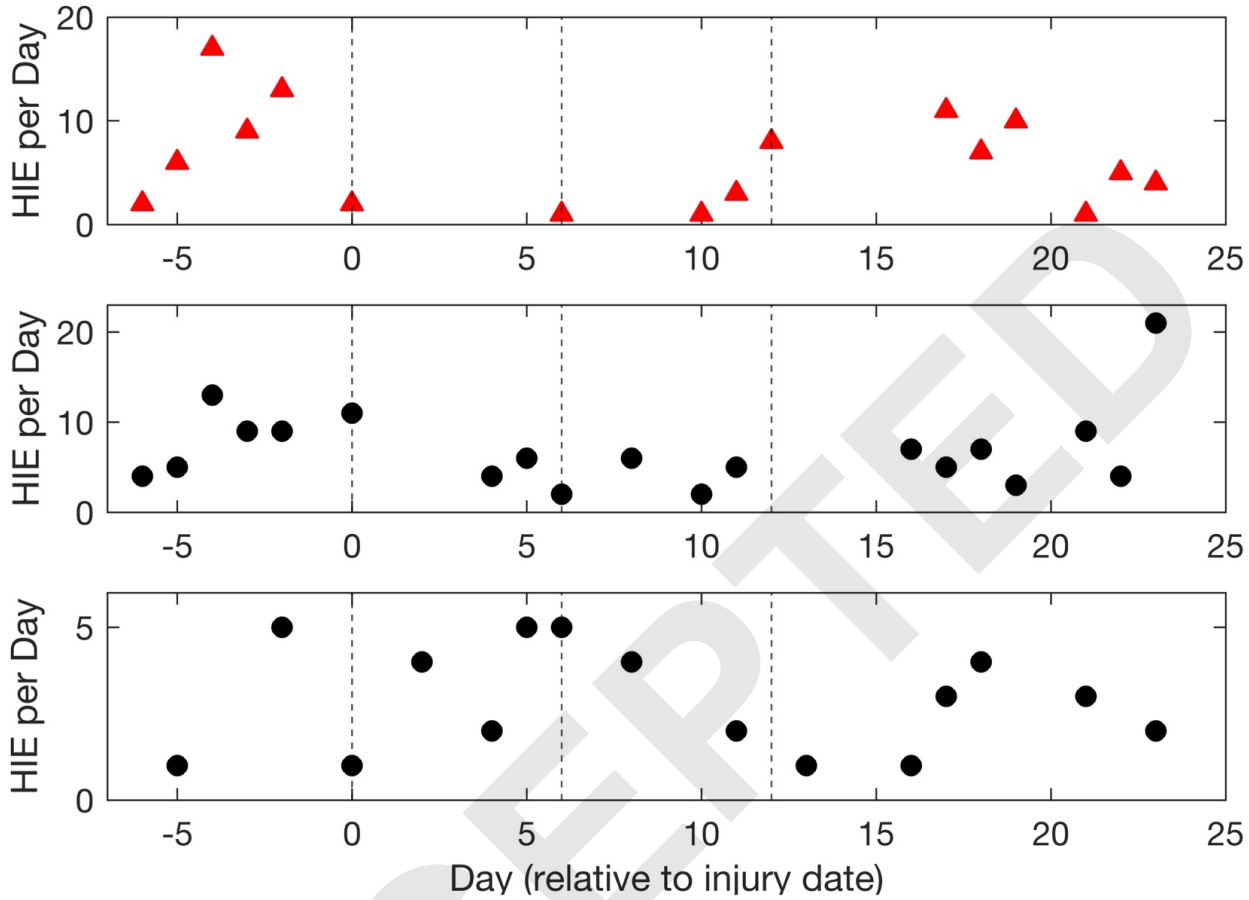
Figure 1



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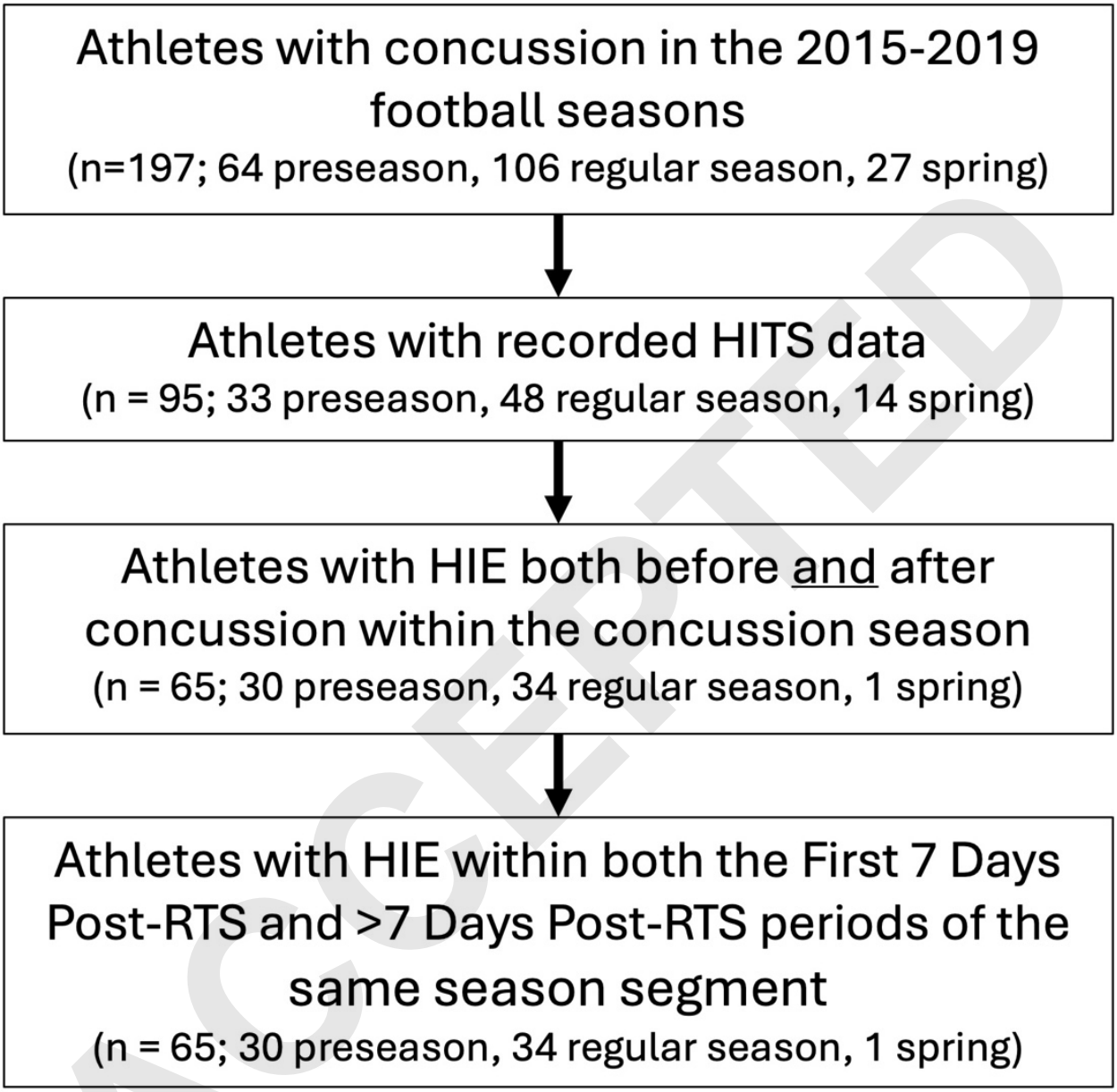

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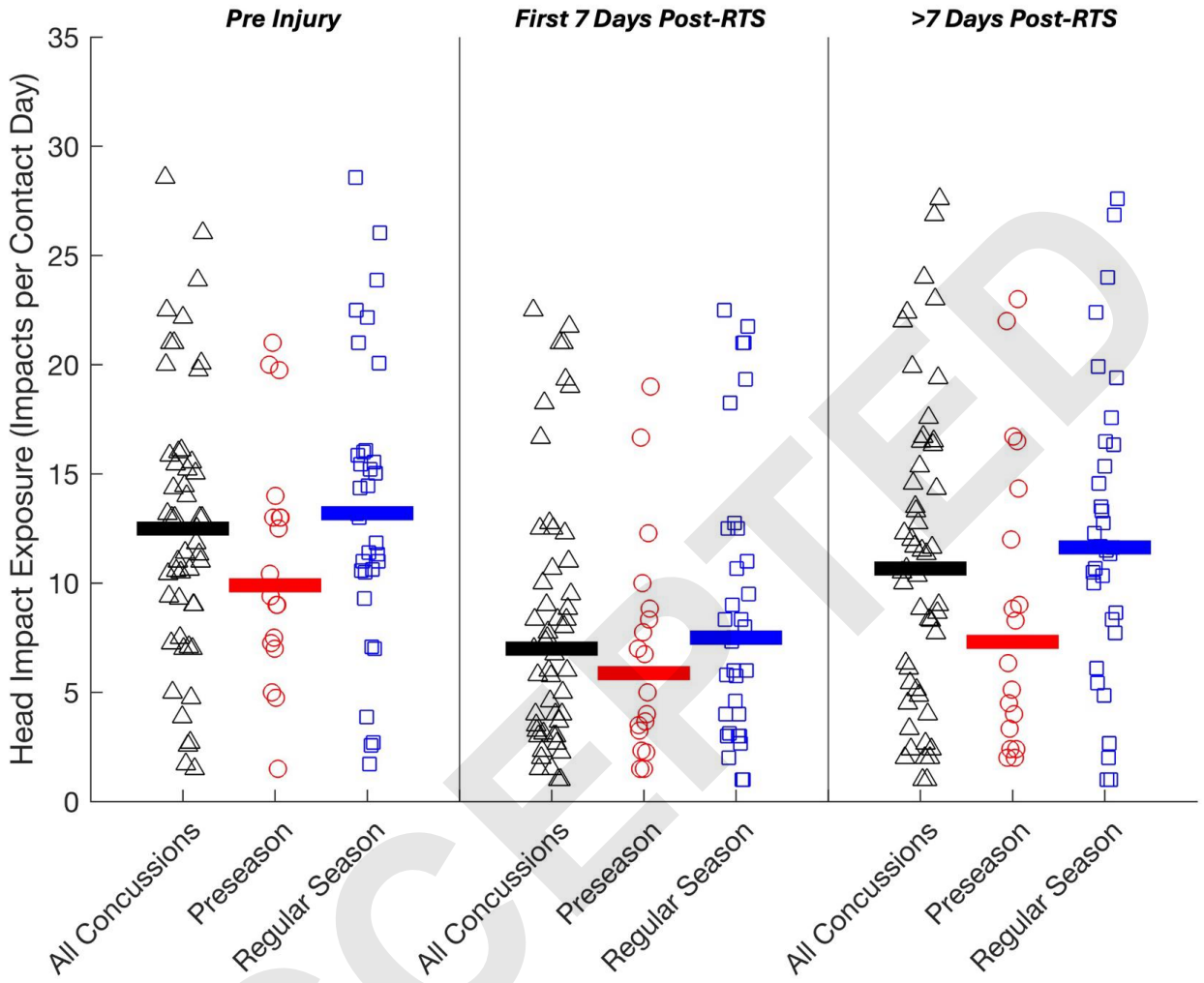

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Figure 4

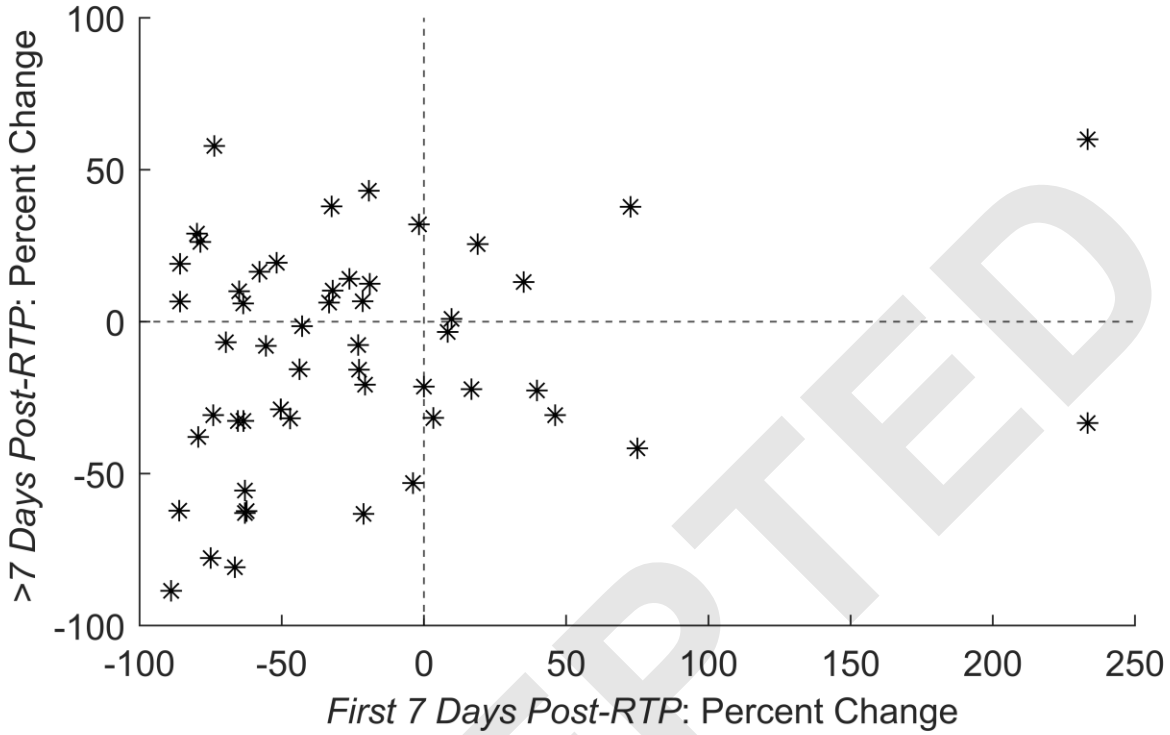

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Figure 5

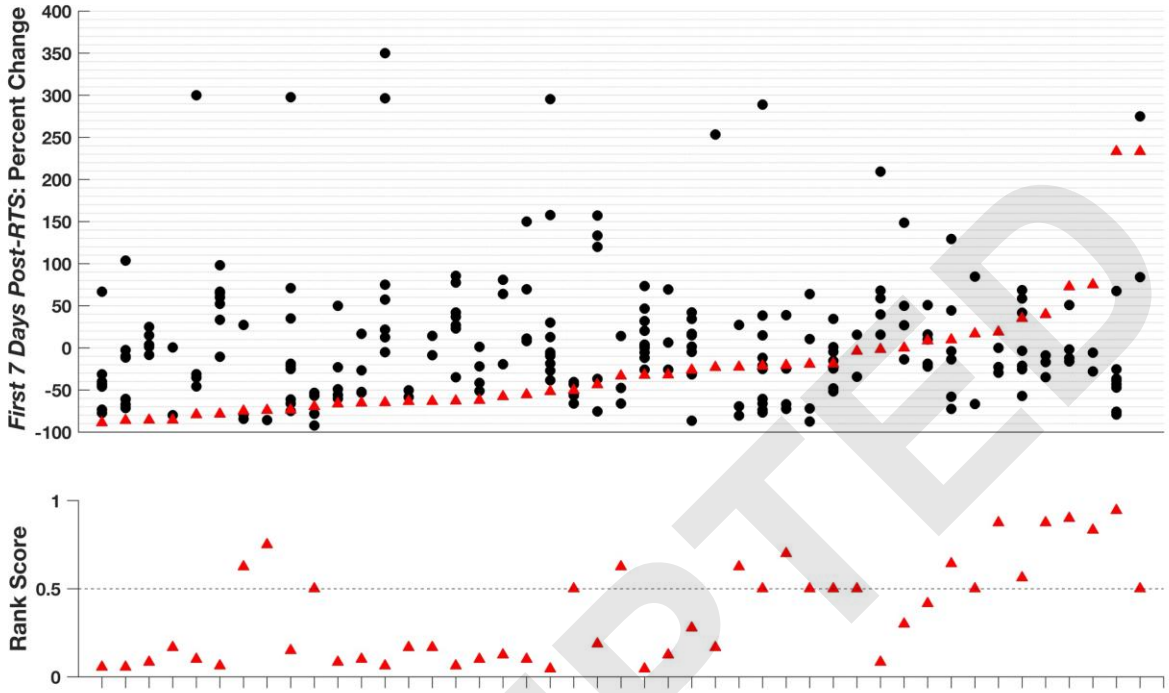

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Figure 6

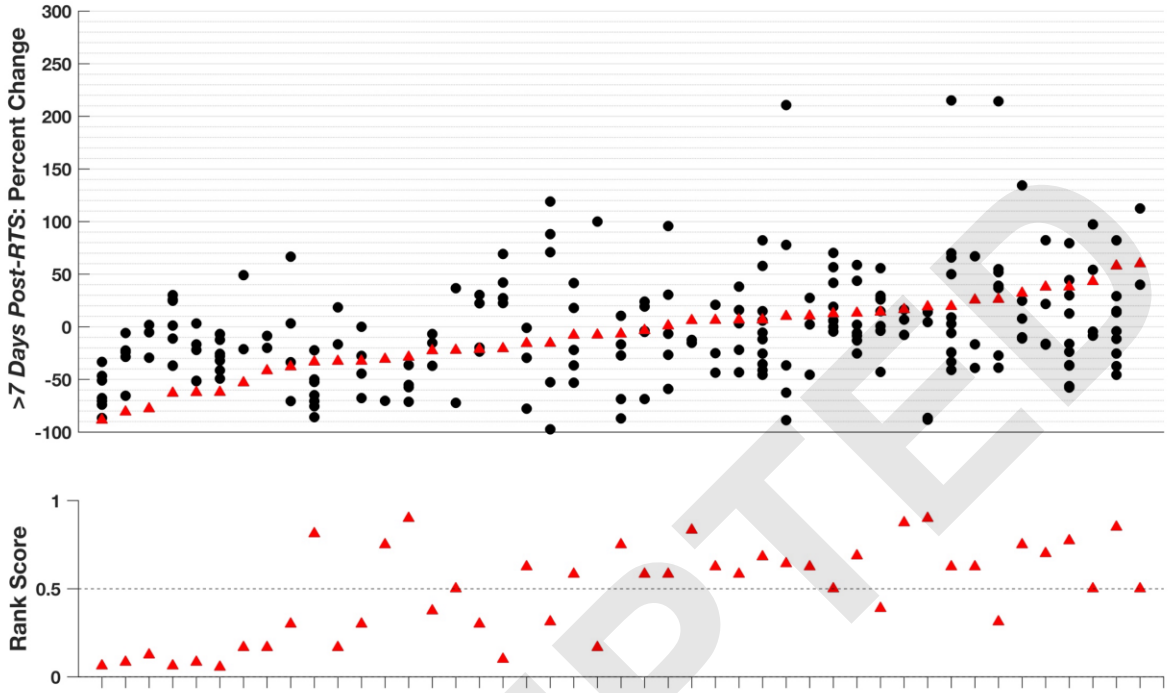

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Figure 7

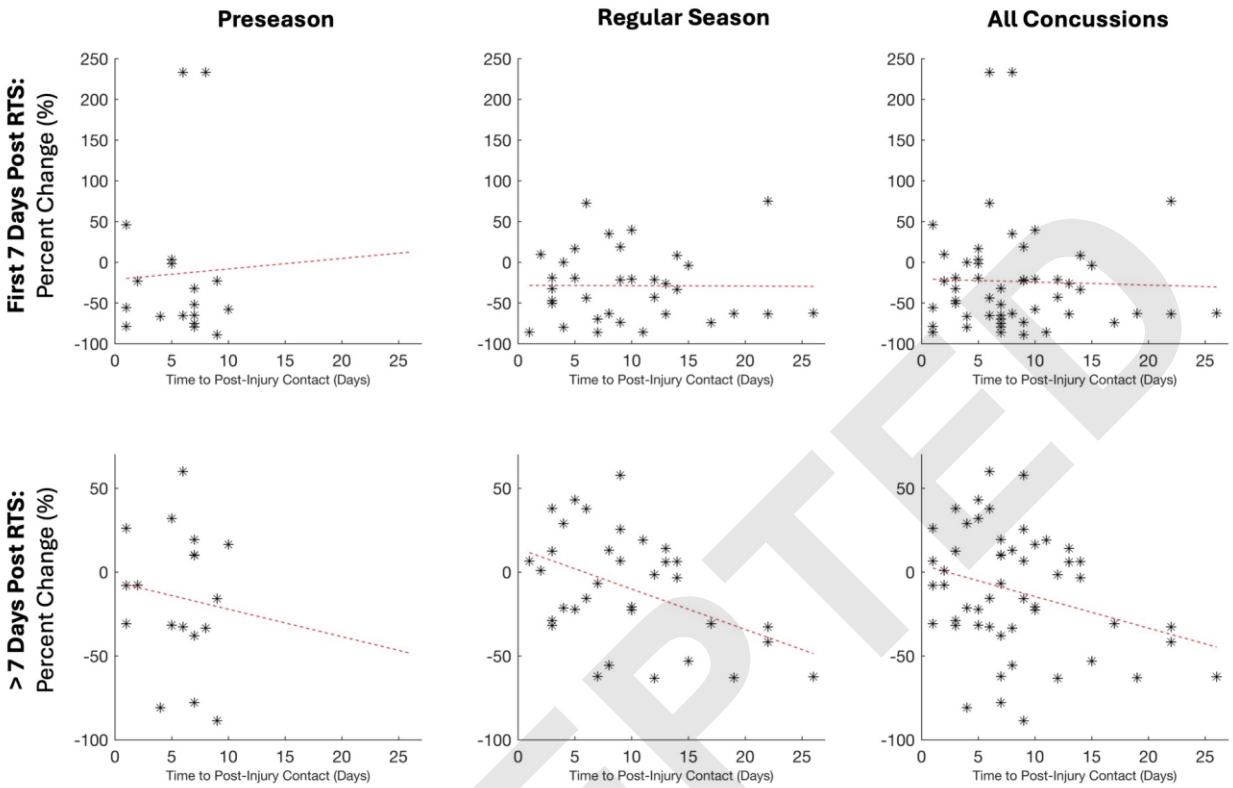

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Table 1: Analysis of player participation prior to concussion and following return-to-sport. Participation was scored as the percentage of offensive or defensive snaps that each athlete participated in relative to their team totals for each game. Post-injury participation change was calculated as the change in post-concussion participation from pre-concussion participation. Post-concussion head impact exposure (HIE) following return-to-sport (both *First 7 Days* and *>7 Days*) is also included.

Athlete	Injury Week	*Pre-Injury* Participation (% of team)	# Weeks *Pre-Injury* Participation	Post-Injury Participation (% of team)	# Weeks Post-Injury Participation	Post-Injury Participation Change	*First 7 Days Post-RTS* HIE Change	*>7 Days Post-RTS* HIE Change
Offense								
3	7	22%	6	29%	5	+7%	-85.7%	+19.0%
10	10	57%	6	21%	3	-36%	-69.6%	-6.7%
23	3	77%	2	91%	10	+14%	-33.3%	+6.3%
26	4	39%	3	50%	9	+11%	-26.2%	+14.1%
30	8	61%	6	62%	4	+1%	-20.7%	-20.7%
31	5	57%	3	55%	8	-2%	-19.4%	+43.1%
32	6	100%	5	100%	7	0%	-19.0%	+12.5%
41	9	27%	7	37%	3	+10%	+39.7%	-22.6%
Defense								
2	8	24%	4	19%	2	-5%	-86.1%	-62.2%
4	6	47%	3	39%	6	-8%	-85.7%	+6.6%
15	3	11%	2	29%	4	+18%	-63.4%	-32.7%
22	2	23%	1	37%	6	+14%	-43.8%	-15.6%
33	3	5%	2	4%	2	-1%	-3.8%	-53.1%
35	4	23%	2	11%	2	-12%	0.0%	-21.8%
39	8	50%	6	88%	3	+38%	+18.9%	+25.6%
42	3	22%	2	10%	5	-12%	+72.6%	+37.8%

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Supplemental Digital Content:

Table S1: Additional data from matched-control analysis, including Position Group, Concussion Season Segment, *Pre-Injury* Daily HIE and number of contact days, Recovery Time, and Post-Injury Percent Change in HIE relative to *Pre-Injury*, Rank Score with higher rank indicating greater reduction in HIE relative to *Pre-Injury*, and Contact Days. Shaded data indicates athletes who were among the top two in largest percent change decrease in HIE when compared against controls.

Athlete Number	Position Group	Concussion Season	*Pre-Injury* Daily HIE	*Pre-Injury* Contact Days	Non-Contact Days following Concussion	*First 7 Days Post-RTS* Percent Change	*First 7 Days Post-RTS* Rank Score	*First 7 Days Post-RTS* Contact Days	*>7 Days Post-RTS* Percent Change	*>7 Days Post-RTS* Rank Score	*>7 Days Post-RTS* Contact Days
1	DB	Preseason	21.0	4	9	-88.9	1 of 9	3	-88.6	1 of 8	5
2	DL	Regular Season	14.4	14	7	-86.1	1 of 9	1	-62.2	1 of 9	7
3	RB	Regular Season	7.0	19	11	-85.7	1 of 3	2	19.0	5 of 5	15
4	DL	Regular Season	21.0	9	1	-85.7	1 of 6	2	6.6	4 of 6	33
5	DB	Preseason	7.3	4	7	-79.3	1 of 5	4	-37.9	2 of 5	8
6	OL	Preseason	7.0	1	1	-78.6	1 of 8	4	26.2	3 of 8	12
7	TE	Preseason	9.0	4	7	-75.0	3 of 4	4	-77.8	1 of 4	5
8	TE	Regular Season	22.2	6	17	-74.1	2 of 2	4	-30.8	2 of 2	26
9	LB	Regular Season	15.2	29	9	-73.7	2 of 10	3	57.9	9 of 10	11
10	OL	Regular Season	13.2	28	7	-69.6	3 of 5	4	-6.7	5 of 6	7
11	DB	Preseason	10.4	7	4	-66.4	1 of 6	2	-80.8	1 of 6	2
12	LB	Preseason	9.4	5	6	-65.4	1 of 5	4	-32.6	2 of 5	6
13	OL	Preseason	20.0	4	7	-65.0	1 of 8	5	10.0	5 of 7	1
14	DL	Regular Season	26.0	30	13	-63.5	1 of 3	2	6.0	3 of 3	10
15	LB	Regular Season	15.9	7	22	-63.4	1 of 3	5	-32.7	1 of 3	31
16	OL	Regular Season	2.7	10	19	-63.0	1 of 8	1	-63.0	1 of 8	4
17	DB	Regular Season	7.1	25	26	-62.3	1 of 5	3	-62.3	1 of 6	3
18	RB	Spring	19.8	4	10	-57.8	1 of 4	3	16.5	4 of 4	3
19	DB	Preseason	9.0	3	1	-55.6	1 of 5	6	-7.9	4 of 6	7
20	OL	Preseason	14.0	4	7	-51.8	1 of 11	4	19.4	8 of 12	7
21	RB	Regular Season	29.0	1	3	-50.3	3 of 5	5	-28.8	5 of 5	33
22	LB	Regular Season	16.0	1	6	-43.8	2 of 8	3	-15.6	3 of 8	14
23	WR	Regular Season	11.0	2	14	-33.3	3 of 4	3	6.3	3 of 4	16
24	OL	Regular Season	11.8	50	3	-32.4	1 of 11	4	37.9	9 of 11	3
25	TE	Preseason	13.0	7	7	-32.1	1 of 4	6	10.3	3 of 4	3
26	OL	Regular Season	14.5	11	13	-26.2	3 of 9	3	14.1	4 of 9	41
27	WR	Preseason	13.0	2	2	-23.1	1 of 3	6	-7.7	1 of 3	8
28	TE	Preseason	4.8	4	9	-22.8	3 of 4	3	-15.8	3 of 4	2
29	LB	Regular Season	10.6	24	9	-21.6	6 of 11	3	6.7	8 of 11	9
30	RB	Regular Season	16.1	26	10	-20.7	4 of 5	4	-20.7	1 of 5	16
31	WR	Regular Season	9.3	10	5	-19.4	3 of 5	2	43.1	3 of 5	13
32	OL	Regular Season	23.9	40	3	-19.0	5 of 9	3	12.5	5 of 9	14
33	LB	Regular Season	13.0	3	15	-3.8	2 of 3	2	-53.1	1 of 3	10
34	LB	Preseason	12.5	4	5	-1.8	1 of 6	7	32.0	5 of 6	2
35	DB	Regular Season	11.0	6	4	0.0	2 of 5	4	-21.8	2 of 5	34
36	DL	Regular Season	20.1	27	14	8.3	3 of 6	4	-3.4	4 of 6	5
37	RB	Regular Season	11.4	38	2	9.7	5 of 7	2	0.9	4 of 6	2
38	WR	Regular Season	2.6	14	5	16.7	2 of 3	1	-22.2	2 of 3	8
39	DB	Regular Season	3.9	23	9	18.9	4 of 4	5	25.6	3 of 4	14
40	DL	Regular Season	15.5	42	8	35.0	5 of 8	5	13.0	6 of 8	7
41	RB	Regular Season	15.0	31	10	39.7	4 of 4	3	-22.6	2 of 4	19
42	DL	Regular Season	10.6	7	6	72.6	5 of 5	4	37.8	4 of 5	23
43	WR	Regular Season	1.7	7	22	75.0	3 of 3	1	-41.7	1 of 3	1
44	WR	Preseason	1.5	2	6	233.3	2 of 3	6	60.0	2 of 3	5
45	DB	Preseason	5.0	2	8	233.3	9 of 9	6	-33.3	7 of 8	3

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